

Oil Tracker: Gulf Flows Recover: Market Fast-Forwards to Future Surplus

- Spot Brent futures prices decreased by 8% over the last week following reported “major progress” regarding US-Iran negotiations, a sizable pick-up in visible Persian Gulf oil flows, a US waiver authorizing Iranian oil sales, and a continuing decline in crude positioning.
- The market is likely extrapolating the swift (thus far) recovery of Mideast supply and already pricing expected future surpluses.
 - Total Gulf exports have returned to 63% of normal levels (7-day moving average) (Exhibit 1) following a rapid recovery of visible Hormuz flows (Exhibit 6), likely reflecting more crossings and a larger share of crossings that are “visible” (ships that are now keeping their AIS on).
 - Empty tanker capacity continued to rise in the Persian Gulf, with 52mb more empty tankers entering the Gulf over the last week vs. 53mb of loadings leaving the Gulf (Exhibit 7).
 - While we don’t expect shipping availability to be a major binding constraint, Iraq has reportedly halted production at the West Qurna 2 oil field on a shortage of empty tankers.
 - There have been no reported attacks on ships in/around Hormuz since June 11.
- The US Treasury issued a 60-day waiver authorizing sales of Iranian oil, including oil transported on previously sanctioned vessels.
 - The waiver may unlock up to around 60mb of Iranian oil on water (Exhibit 2).
 - However, we do not expect a large pick-up in Iranian oil production even if the sanctions relief extends beyond the August 21st expiration date.
 - On the demand side, Asia and China in particular will likely remain the main buyers of Iranian crude (Exhibit 2) as vast EU and UK sanctions on Iranian oil and vessels remain in place.
 - On the supply side, Iran grew its production by 1.3mb/d (59%) between 2020-2025 even under the US “maximum pressure” campaign, suggesting that US sanctions might not be a very tight constraint on production growth.
- The pace of global visible stocks draws has slowed sharply in June to -1.8mb/d from -5.5mb/d in May (Exhibit 10) as oil on water has picked up.

Yulia Zhestkova Grigsby

+1(646)446-3905 |
yulia.grigsby@gs.com
Goldman Sachs & Co. LLC

Alexandra Paulus

+1(212)902-7111 |
alexandra.paulus@gs.com
Goldman Sachs & Co. LLC

Filippo Cuscito

+44(20)7051-9073 |
filippo.cuscito@gs.com
Goldman Sachs International

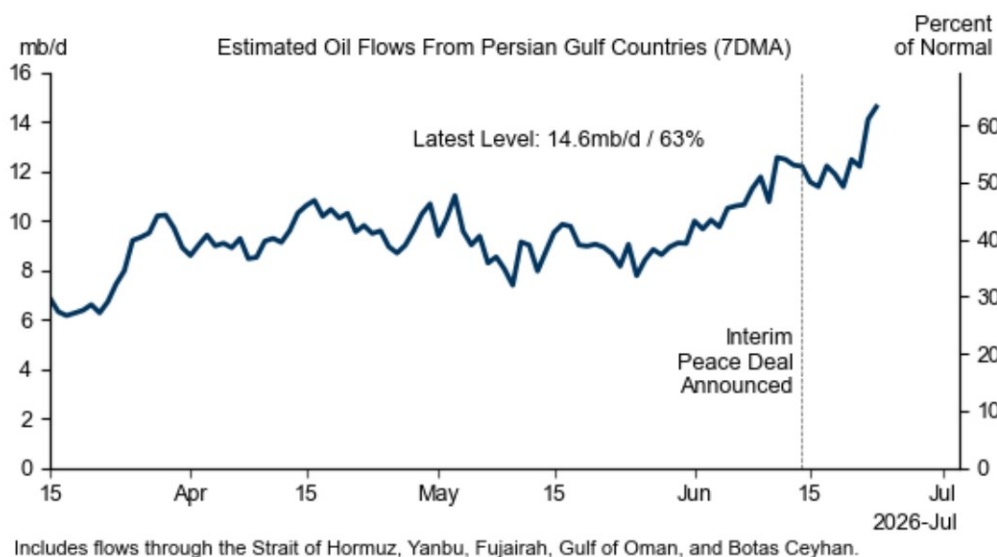
Daan Struyven

+1(212)357-4172 |
daan.struyven@gs.com
Goldman Sachs & Co. LLC

- Global oil in transit has increased by 137mb since late March, offsetting nearly 60% of the initial drop as global exports have ramped up and demand has likely fallen ([Exhibit 3](#)).
 - US exports accounted for all of the pick-up in oil on water in April-May, while Persian Gulf ex Iran exports drove nearly all of the June recovery.
- Some of the demand losses may prove sticky, especially in Asia, as:
 - China crude imports continue to decline and are now down 4.5mb/d YoY despite the moderation in prices.
 - EV sales have accelerated since the beginning of the Hormuz shock, with the EV penetration rate up 3.4pp globally and 11.4pp in China.
- The market may be placing an increasing probability on our oil price downside scenario, which features Brent around \$60/bbl by Dec26.
 - Aside from potentially larger demand scarring, Middle East supply may recover even faster than we expect (for example, UAE exports are already reportedly at 85% of pre-war levels), while production beats outside of the Middle East may continue.
 - High supply expectations have flipped Dubai prompt timespreads into contango ([Exhibit 17](#)).
 - However, the US oil market is tight following record exports, in contrast to March when US stocks were building.
 - Last week, total US oil inventories hit their lowest level since 1984, while Cushing crude storage dropped below 19mb.
- Beyond the spot-price selloff, the market is increasingly challenging its prior assumption that long-dated prices need to incorporate a sticky security premium.
 - While supply disruption risk is high, the ability of the oil market (via elasticity of demand and flexibility of Mideast supply channels) to respond to the sharpest oil supply shock ever has been a key surprise that is likely eroding this security premium.

Charts of the Week

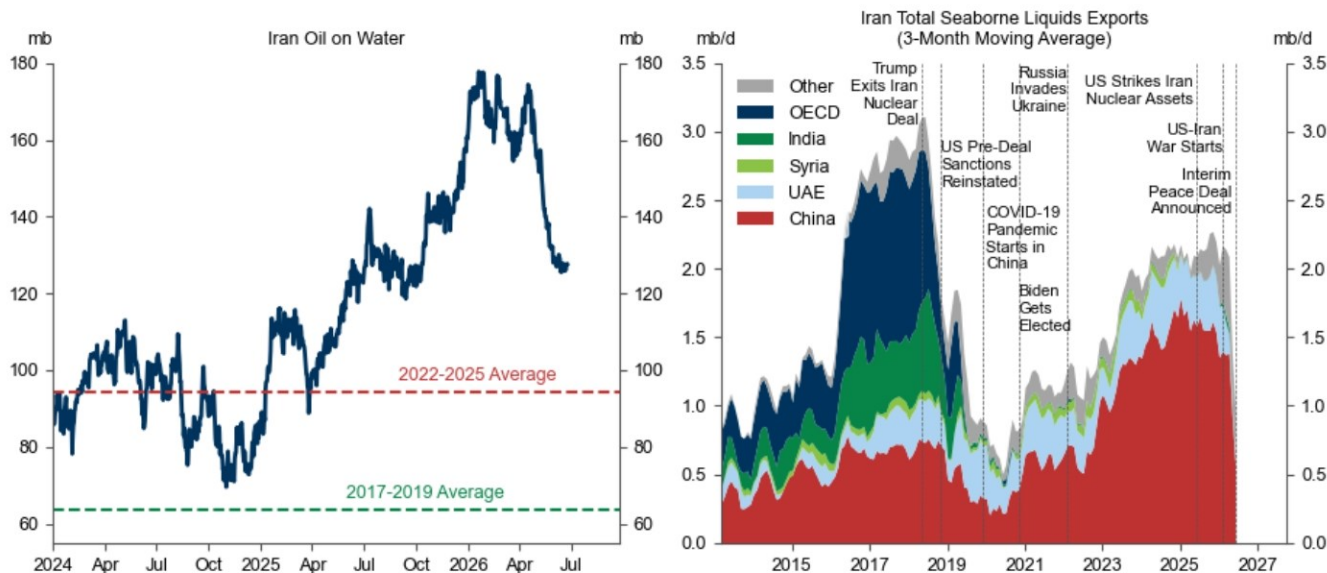
Exhibit 1: Oil Exports from the Persian Gulf Have Returned to 63% of Normal Levels



Yanbu is on the west coast of Saudi Arabia, bordering the Red Sea and connected to Saudi eastern oil fields via the East-West pipeline. Fujairah lies east (outside) of the Strait of Hormuz and is connected to Abu Dhabi's onshore fields via the Abu Dhabi Crude Oil Pipeline (ADCOP). The Gulf of Oman connects the Strait to the Arabian Sea (southeast). The Kirkuk-Ceyhan pipeline allows northern Iraqi crude to be pumped through Turkey to the Mediterranean sea. "Normal" flows are assumed to be 20mb/d for Strait of Hormuz and 2025 average for Yanbu (1.4mb/d), Fujairah (1.7mb/d), Gulf of Oman (0mb/d), and Botas Ceyhan (0mb/d).

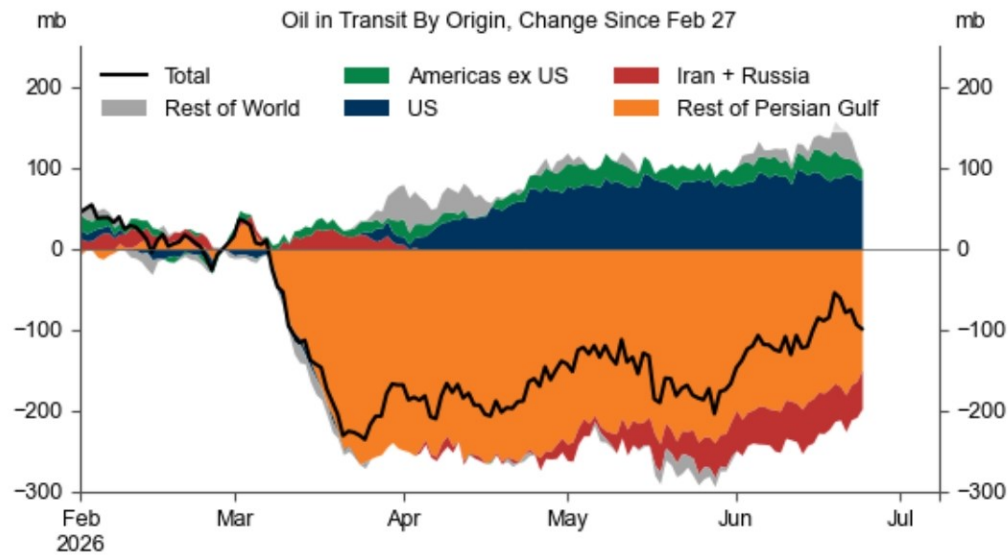
Source: Kpler, S&P Global Commodities at Sea, Goldman Sachs Global Investment Research

Exhibit 2: The US 60-Day Waiver May Unlock Up to Around 60mb of Iran Oil on Water Overhang



Source: Kpler, Goldman Sachs Global Investment Research

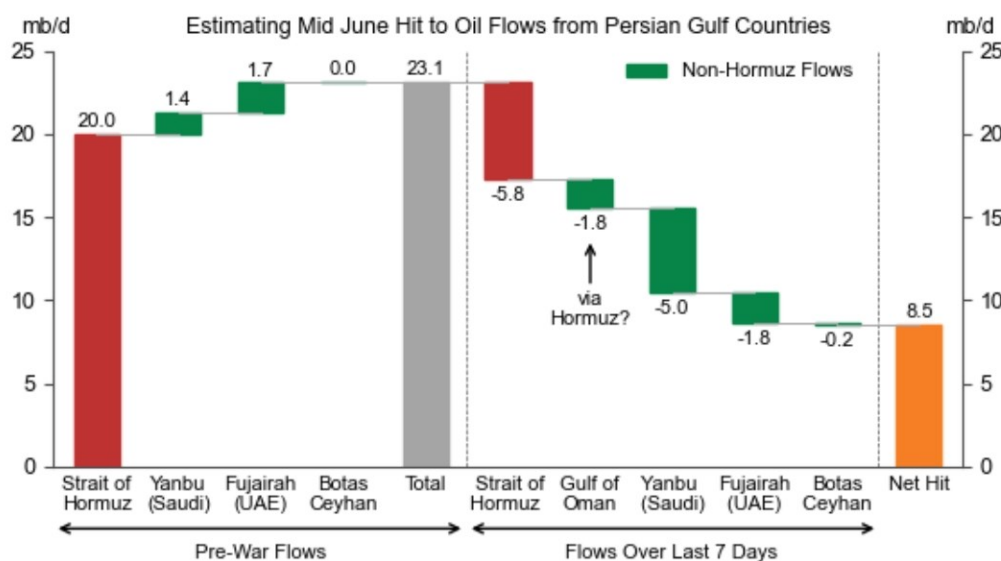
Exhibit 3: Oil in Transit Has Increased by 68mb Since May 31 as Persian Gulf Flows Are Recovering



Source: Kpler, Goldman Sachs Global Investment Research

1) Persian Gulf Exports

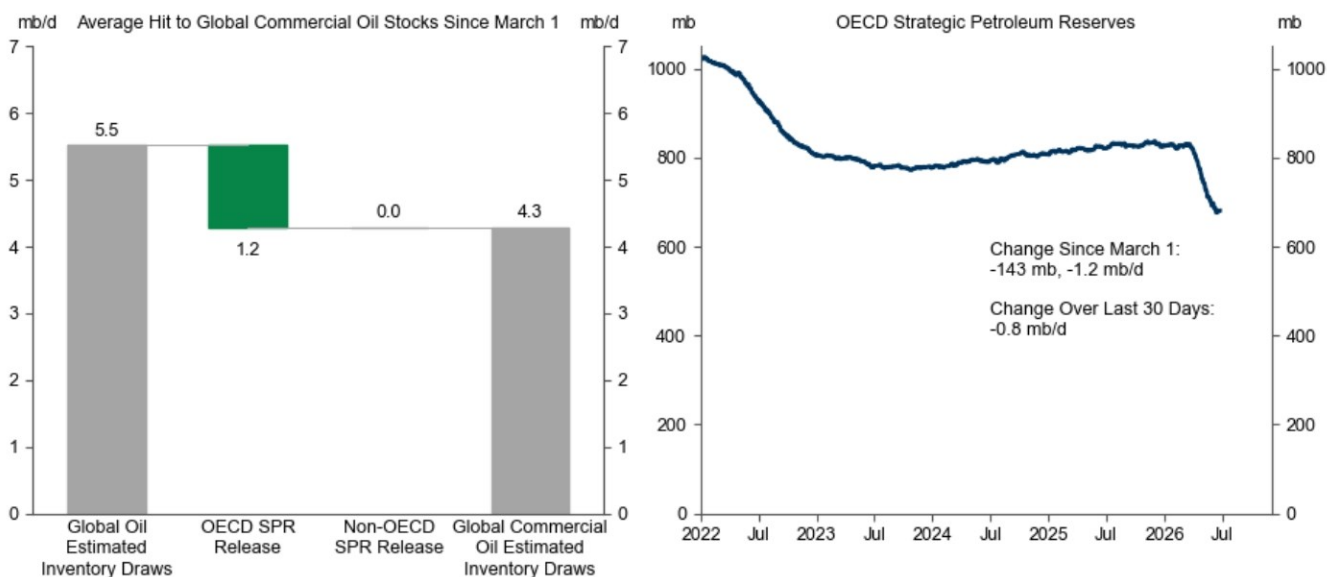
Exhibit 4: Normalization in Oil Exports From Gulf Producers to Their Pre-War Level May Be Achieved With an 8.5mb/d Increase in Hormuz Flows From Current Levels



Yanbu is on the west coast of Saudi Arabia, bordering the Red Sea and connected to Saudi eastern oil fields via the East-West pipeline. Fujairah lies east (outside) of the Strait of Hormuz and is connected to Abu Dhabi's onshore fields via the Abu Dhabi Crude Oil Pipeline (ADCOP). The Gulf of Oman connects the Strait to the Arabian Sea (southeast). The Kirkuk-Ceyhan pipeline allows northern Iraqi crude to be pumped through Turkey to the Mediterranean sea.

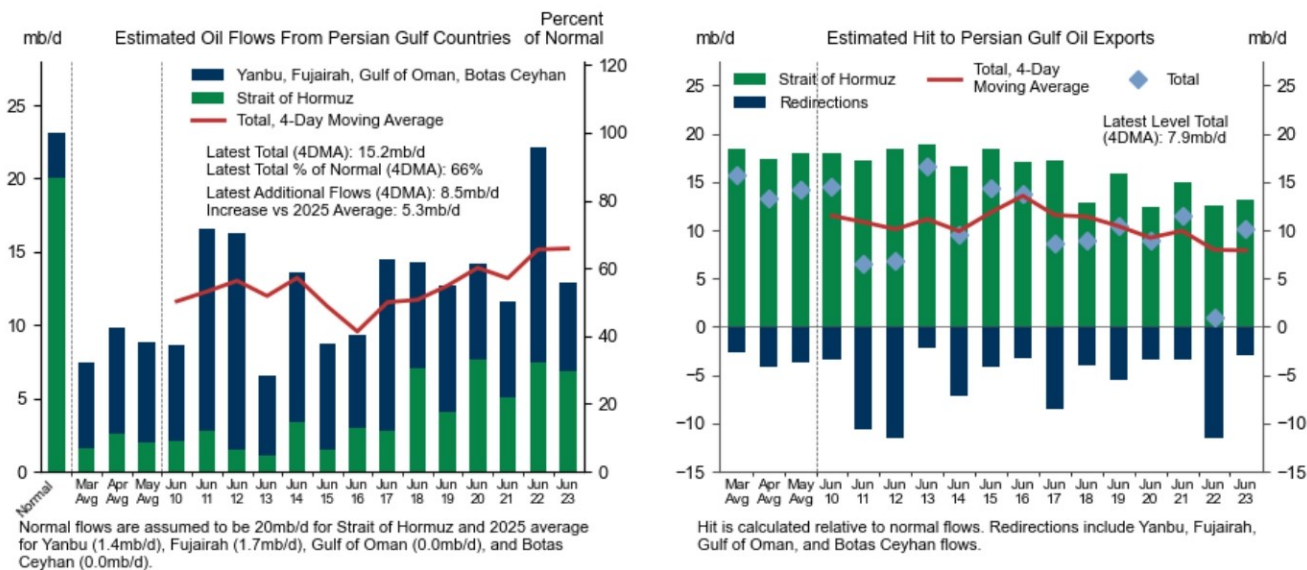
Source: Kpler, S&P Global Commodities at Sea, Goldman Sachs Global Investment Research

Exhibit 5: We Estimate That 1.2mb/d of Visible OECD SPR Releases Have Reduced the Estimated Hit to Global Commercial Oil Stocks Since March to 4.3mb/d

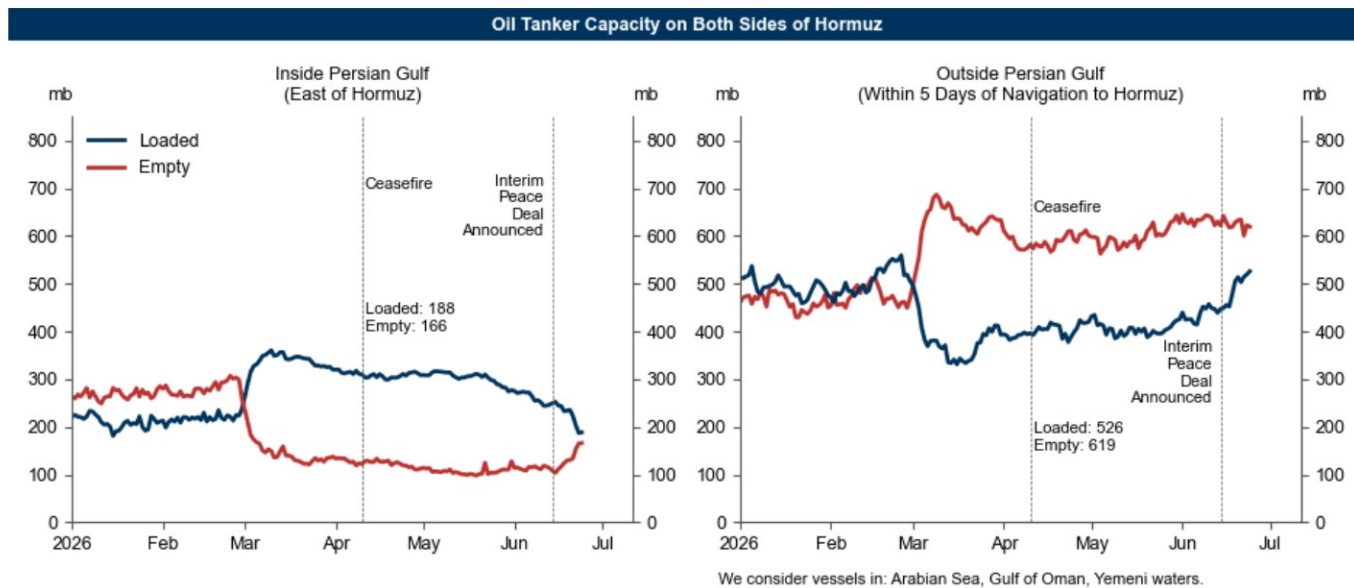


We estimate global oil inventory draws from latest GS oil balance.

Source: Kpler, Goldman Sachs Global Investment Research

Exhibit 6: The Estimated Total Hit to Oil Flows from the Persian Gulf Decreased to 7.9mb/d (4-Day Moving Average)

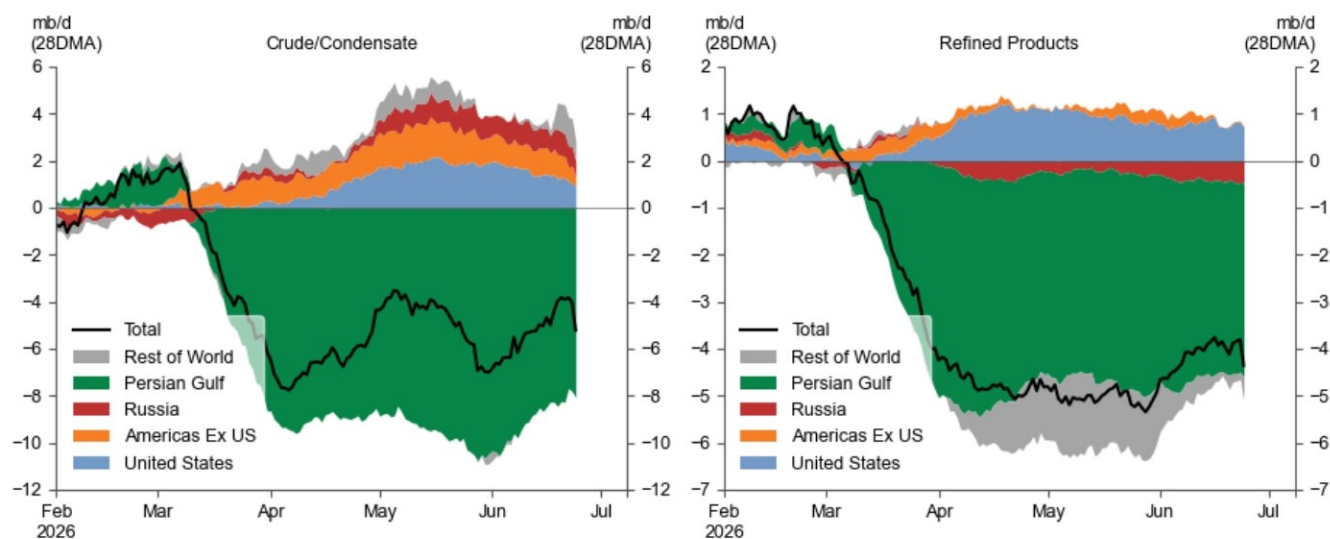
Source: S&P Global Commodities at Sea, Kpler, Goldman Sachs Global Investment Research

Exhibit 7: We Estimate That Empty Oil Tankers Within the Strait or Within 5 Days of Navigation Have the Capacity to Load 785mb

Source: Kpler, Goldman Sachs Global Investment Research

Exhibit 8: 35% of the Hit to Persian Gulf Crude/Condensate Exports Is Currently Being Offset, With Contributions From Higher Exports from the US/Russia/Americas Ex US of 11pp/8pp/5pp

Global Oil Exports vs. 2025 Average

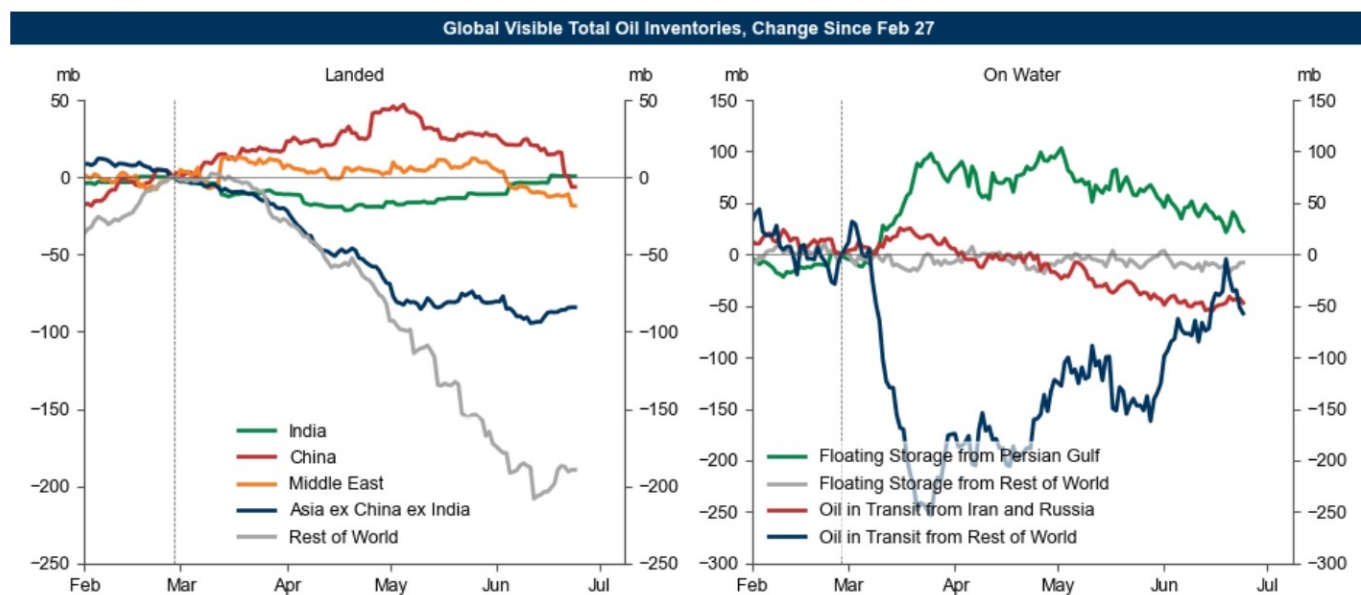


Refined products include LPG.

Source: Kpler, Goldman Sachs Global Investment Research

2) Inventories

Exhibit 9: China and Middle East Landed Visible Inventories Have Drawn by 2.2mb/d and 0.8mb/d, Respectively, Over the Last 14 Days



Source: IEA, Kpler, DOE, Euroilstocks, ARA PJK, PAJ, Haver, Goldman Sachs Global Investment Research

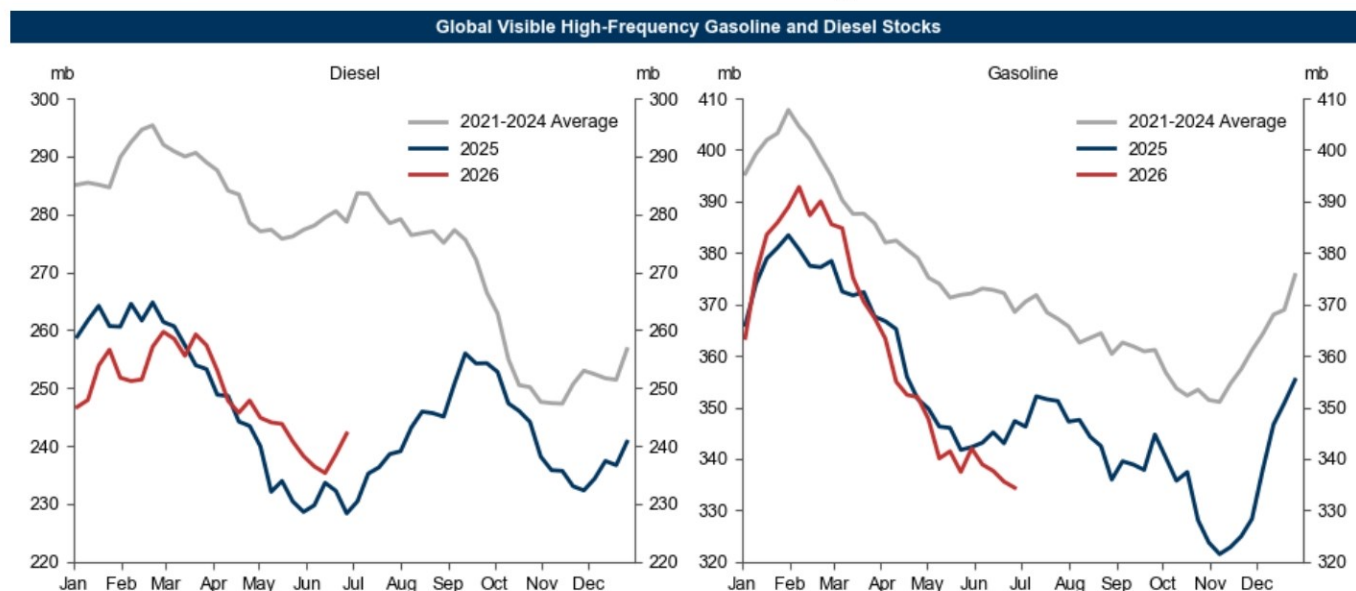
Exhibit 10: Global Visible Draws Have Averaged 3.4mb/d Since March 1st

Visible Oil Stocks, Month-Over-Month Changes (mb/d)				
	June	May	April	Average Since March 1st
Global Visible Stocks	-1.8	-5.5	-1.4	-3.4
Landed Crude Stocks	-2.5	-2.9	-1.8	-1.7
OECD	-1.3	-2.7	-2.1	-1.5
China	-1.1	-0.3	0.4	-0.1
Non-OECD Ex China	-0.1	0.1	0.0	-0.1
Landed Products Stocks	-0.3	-0.7	-1.1	-0.9
OECD NGL	0.1	0.2	0.4	0.2
OECD Refined Products	0.2	-0.5	-1.5	-0.8
Non-OECD Total Products	-0.6	-0.3	0.1	-0.2
Oil on Water	1.0	-1.9	1.4	-0.9
Floating Crude	-1.3	-0.9	0.4	0.1
Floating Products	-0.4	0.0	0.0	0.1
Crude in Transit	3.3	-0.5	1.4	-0.2
Products in Transit	-0.6	-0.5	-0.4	-0.8

Latest observation is today, and data is unsmoothed. "Crude" categories (landed and on water) include both crude and condensate. Month-over-month changes are calculated by taking the end-of-month level, subtracting the previous month's end-of-month level, and dividing by the number of days.

Source: Goldman Sachs Global Investment Research

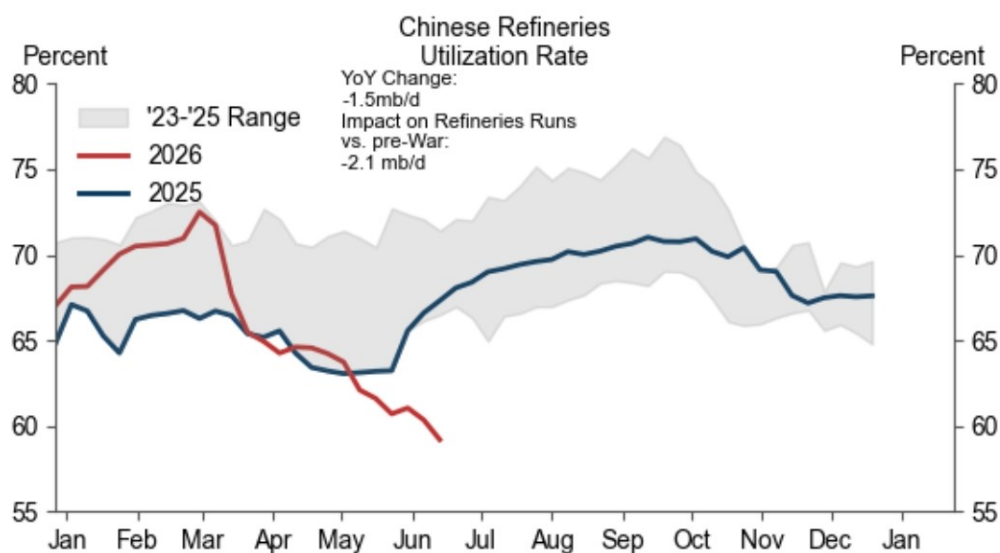
Exhibit 11: Global Diesel High-Frequency Visible Stocks Increased to 6% Above Their Year-Ago Levels Last Week, While Gasoline Stocks Continued to Trend Downwards to 4% Below Their Year-Ago Levels



Source: DOE, Longzhong, ARA PJK, Fujairah, Singapore Enterprise, IEA, PAJ, Goldman Sachs Global Investment Research

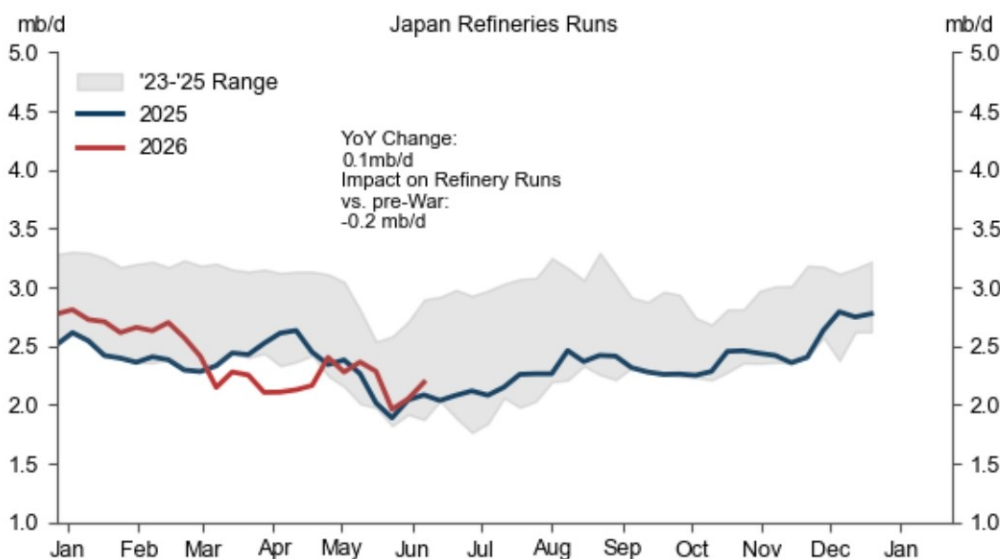
3) Refining

Exhibit 12: Estimated Chinese Refineries Runs Have Declined by 1.5mb/d From a Year Ago



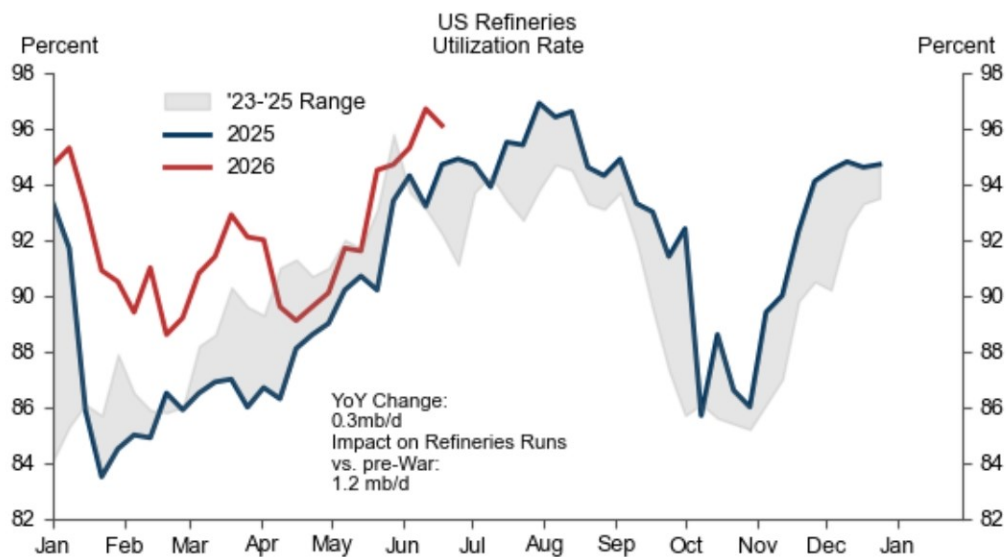
Source: Oilchem, Goldman Sachs Global Investment Research

Exhibit 13: Japanese Refineries Runs Are Starting to Pick Up



Source: PAJ, Goldman Sachs Global Investment Research

Exhibit 14: The Utilization Rate of US Refineries Is Up 0.3mb/d Year-Over-Year as US Product Margins Remain High

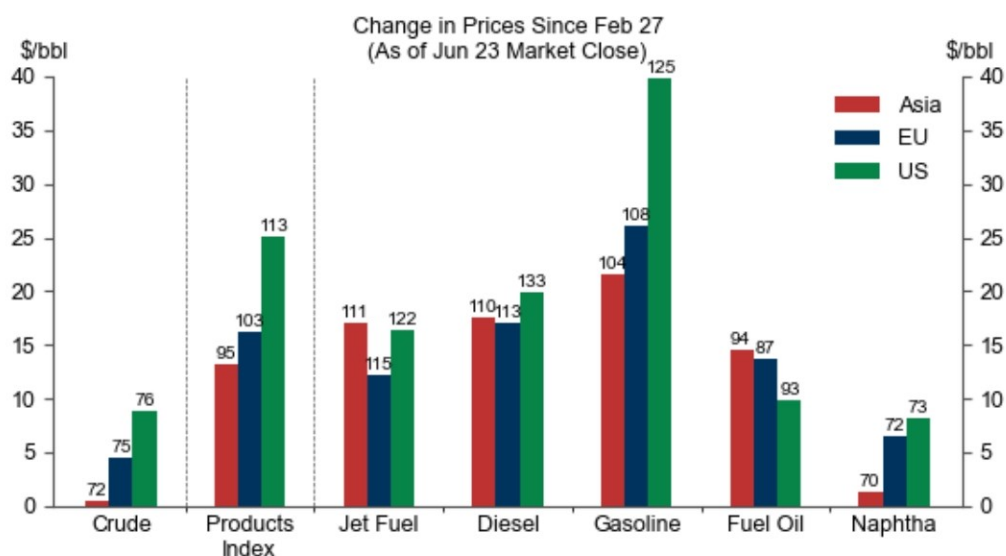


Source: EIA, Goldman Sachs Global Investment Research

4) Energy Prices

Energy Prices Across Regions

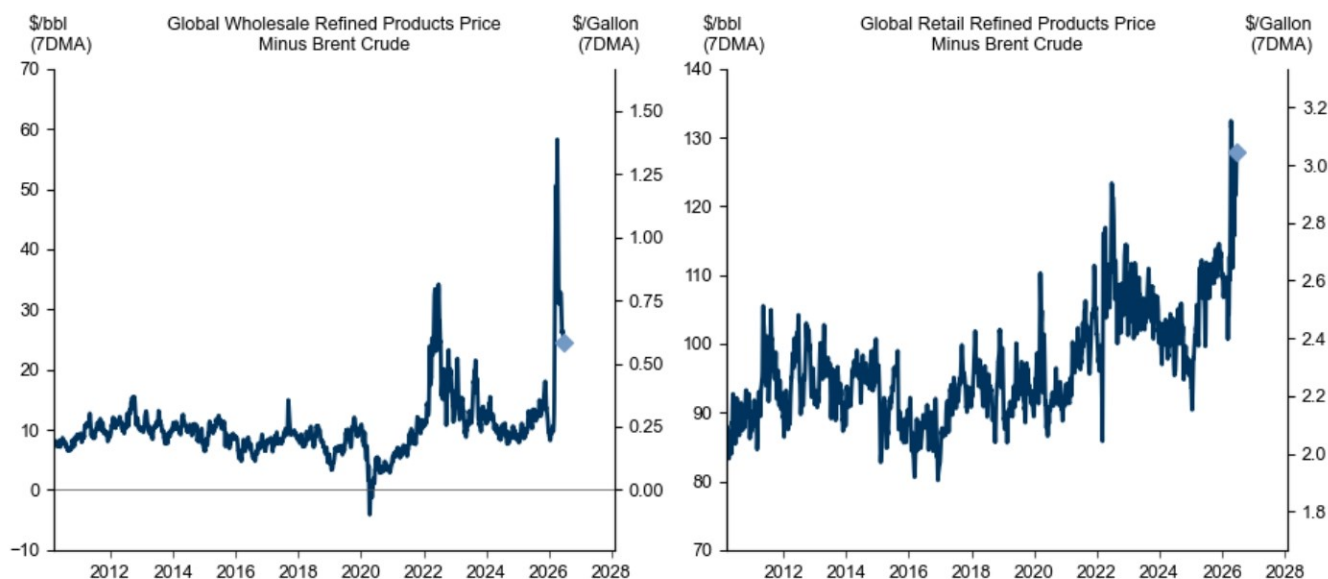
Exhibit 15: US Wholesale Gasoline Prices Continue to Outperform Regional Counterparts and Products Across the Barrel on Strong Summer Demand and Low Stocks



Bar labels indicate crude/refined products prices as of June 23rd market close. We consider cash or cash-equivalent prices.

Source: ICE, Platts, Goldman Sachs Global Investment Research

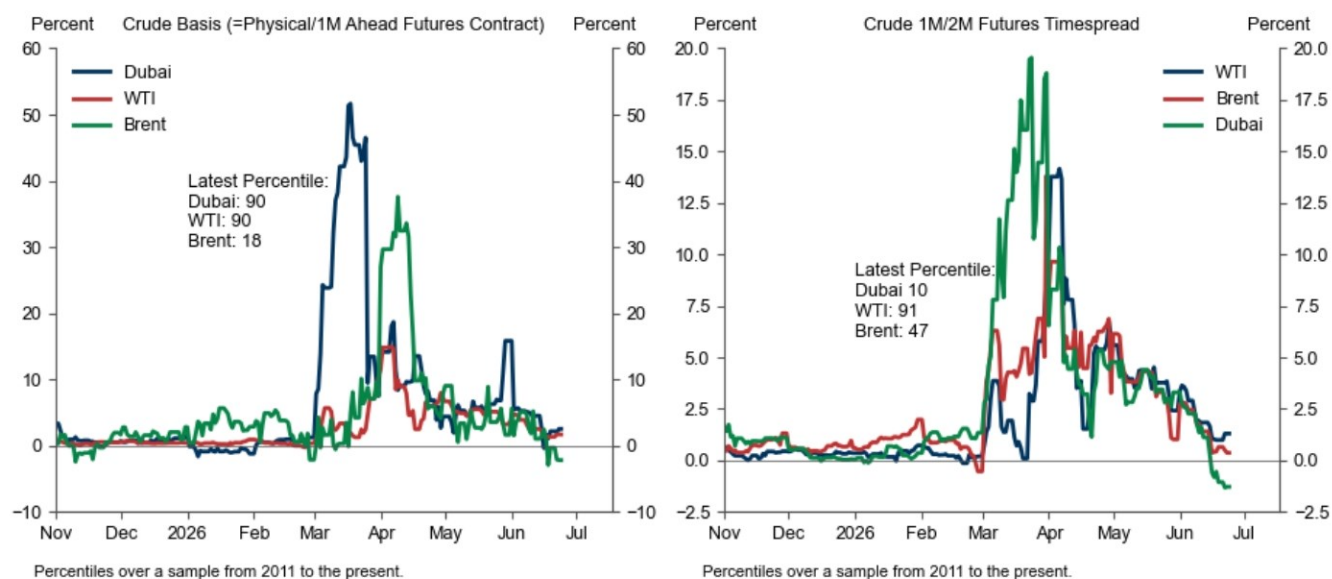
Exhibit 16: Global Wholesale Refined Products Margins Continue to Normalize, While Retail Margins Remain Near Their All-Time High



The global wholesale refined product price is a demand-weighted average of wholesale price indices in the US, Europe, and Asia, which are themselves demand-weighted averages of gasoline, diesel, jet fuel, naphtha, and fuel oil prices. The global retail refined product price is a simple average of global retail gasoline and diesel price indices, which are each demand-weighted averages of 40 country-level retail product prices.

Source: Platts, EIA, Government Sources, ICE, Goldman Sachs Global Investment Research

Crude Prices

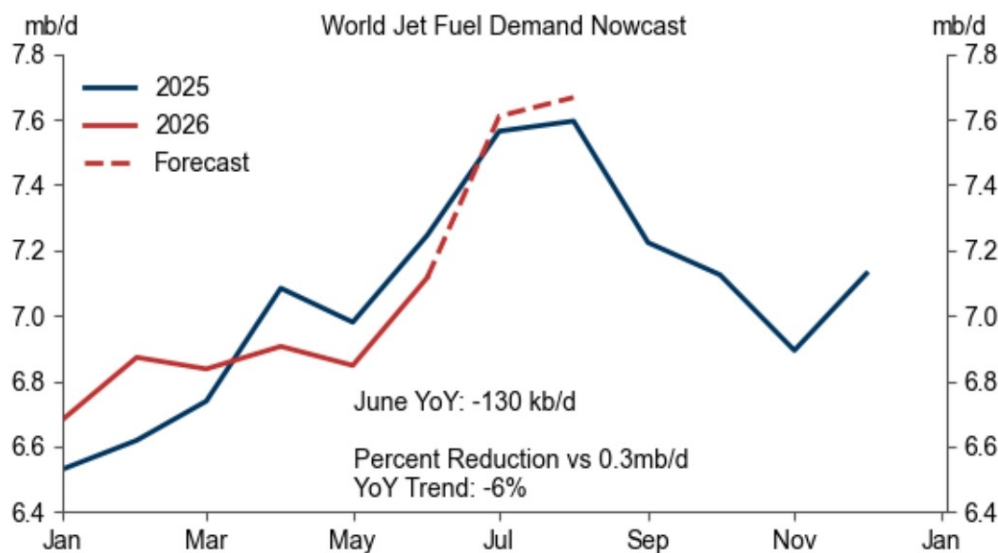
Exhibit 17: Dubai Prompt Timespreads Remain in Contango, While the Brent Physical Contract Is Now Trading at a Discount to Its Financial Counterpart

Basis represents the difference (in % of 1M ahead futures) between spot physical crude futures and 1M ahead financial futures. We include Brent, Dubai, and WTI in crude basis on the left panel and in 1M/2M timespreads on the right panel.

Source: Platts, CME, Goldman Sachs Global Investment Research

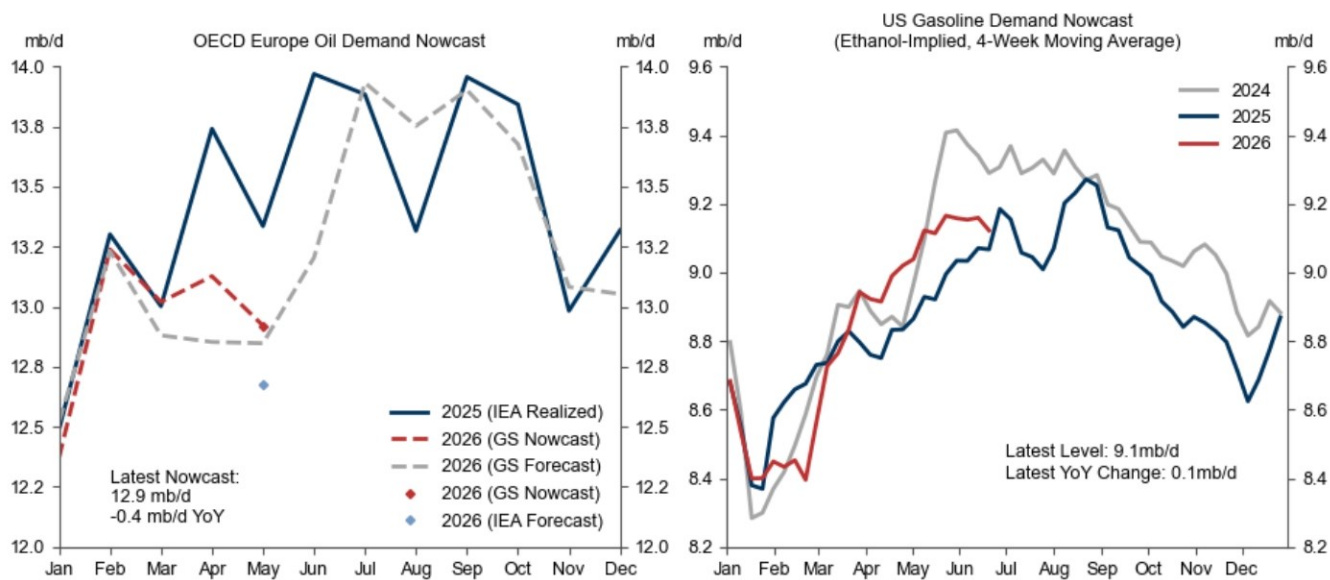
5) Demand

Exhibit 18: We Estimate That Global Jet Fuel Demand in June Will be 130kb/d Softer Year-Over-Year, or 6% Below Trend



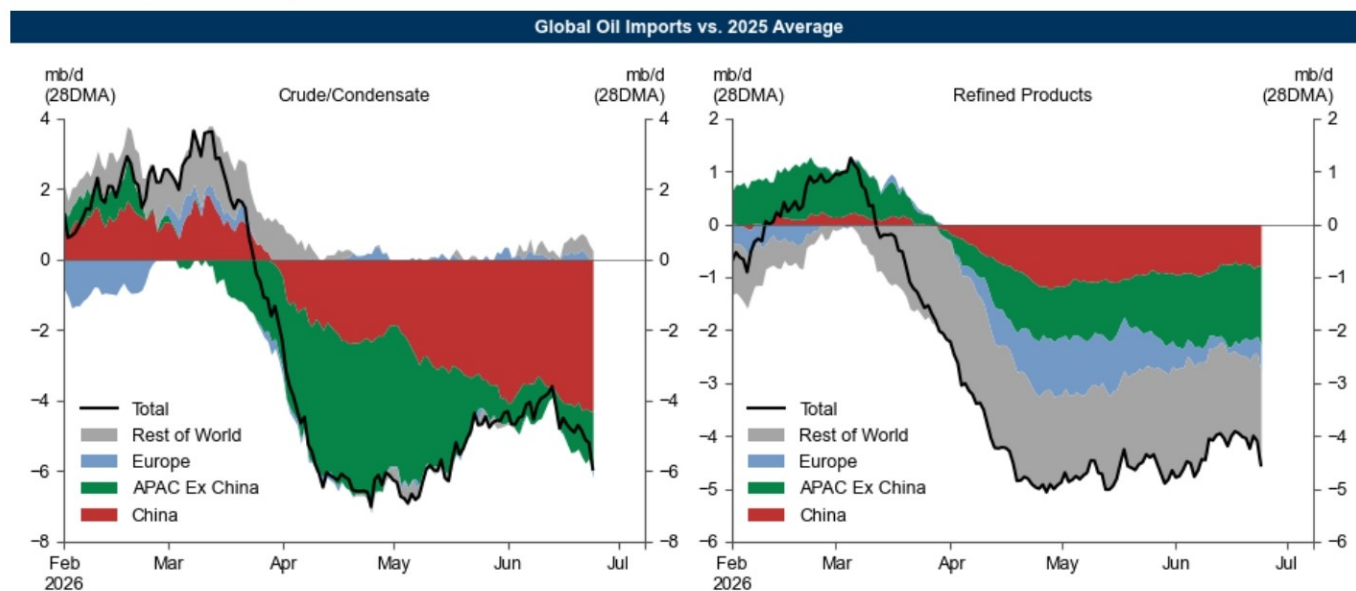
Source: EIA, OAG, Goldman Sachs Global Investment Research

Exhibit 19: Our US Gasoline Demand Nowcast Is Now Roughly In Line With Its Year-Ago Seasonal Levels



Source: IEA, S&P, Kpler, GTT, Oilchem, MySteel, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 20: Global Crude/Condensate Imports Are Down 5.9mb/d from 2025 Average Levels, and Refined Products Imports Are Down 4.6mb/d



Refined products include LPG.

Source: Kpler, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Yulia Zhestkova Grigsby, Alexandra Paulus, Filippo Cuscito and Daan Struyven, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Yulia Zhestkova Grigsby Goldman Sachs & Co. LLC, Alexandra Paulus Goldman Sachs & Co. LLC, Filippo Cuscito Goldman Sachs International, Daan Struyven Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Disclosure

Iran is subject to comprehensive sanctions by the United States.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors

should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.